WHAT IS THE CHANGE I WANT TO CREATE?

- Barcelona will replace thousands of plane trees. It surely has criteria for the new species, but they are not published, and no public document says what a choice means for heat or flooding.
- For every tree to replace, designers and the city should see which species gives more summer shade and holds back more rain at that spot.
- Always within the city's rules: no species above 15% of the trees, and bigger trees only on wider sidewalks.

OUTPUT

- Shade & Sponge: an open-source tool (Python core, web app, Grasshopper link).
- In: tree inventory, LiDAR crowns, terrain, design storm, summer weather file.
- Out: a species and position for each replacement, scored on summer shade / UTCI and runoff against the city's own replacement mix.
- Tested in Porta (Nou Barris); run on 33 neighbourhoods.

SHORT-TERM OUTCOMES

- Technological: a reproducible method and public code (MIT), with a heat check in Ladybug and Honeybee.
- Environmental (simulated): more summer shade than the city's real replacement mix in all 33 neighbourhoods. Ladybug and Honeybee show pedestrian heat stays as with today's planes (within ±0.2 °C UTCI), so the gains are in shade, runoff, species diversity and winter sun.
- Knowledge: species and positions chosen against pedestrian heat and runoff together, which the literature reviewed does not do yet.

MEDIUM-TERM OUTCOMES

- Technological: designers and Parcs i Jardins test it on real replacement campaigns or pilot streets.
- Replication: transfer to a second Mediterranean city (Florence or Mataró).
- Social: streets keep their shade and comfort while the planes are replaced, starting where heat and flooding overlap.
- Economic: fewer trial-and-error plantings, faster design.

LONG-TERM IMPACT

- Environmental: canopy moves toward the city's 30% target for 2037 with no species above 15%; less runoff to sewers in flood hotspots.
- Social: no loss of pedestrian comfort during the plane replacement, with more canopy where it is lacking.
- Economic: avoided drainage and heat-health costs.
- A replicable method for cities replacing monocultures.

ASSUMPTIONS

- The city keeps replacing plane trees over the next decade (press, 2026: more than half).
- A fast shade proxy and a simple runoff model rank species correctly. Partly checked for heat: proxy shade tracks Ladybug UTCI ($r = -0.82$).
- LiDAR crowns and LAI represent mature trees; light through a crown depends on its LAI.
- The city keeps its current rules (no species above 15%, tree size set by sidewalk width): the tool is built on them.
- The city and designers are willing to use an open tool.

RISKS

- Optimising two services can concentrate on a few 'best' species: pest risk (mitigated by the cap and a genus check).
- Results over-trusted: simulated, not measured. The heat difference is small, and its sign flips between solid and porous crowns.
- Big crowns vs winter sun, façades, roots, maintenance, allergies.
- Benefits arrive only when trees mature (decades).
- Used to justify removing healthy trees.

POTENTIAL INDICATORS

- % sidewalk in summer shade vs city mix (pts)
- Δ UTCI, 12–17 h, July day (°C; Ladybug, Honeybee)
- Δ runoff, design storm (m^3, %)
- Agreement: proxy vs Ladybug (r), Python vs Grasshopper (r)
- Neighbourhoods where the tool beats the city mix (n / 33)

POTENTIAL INDICATORS

- Streets or projects that use the tool (n)
- Design time per street, before vs after (h)
- Measured air / surface temperature in pilot streets
- Result quality in the transfer city
- Designer feedback (interviews, ease of use)

POTENTIAL INDICATORS

- Canopy cover per neighbourhood (LiDAR; Porta 20.1% in 2021 vs 30% target)
- Share of the largest species (planes ~27% → ≤15%)
- Hours above strong heat stress (UTCI > 32 °C)
- Sewer overflows and flood reports in hotspot streets
- Cities that adopt the method

Shade & Sponge · Theory of Change choosing replacement street trees for shade and runoff (Barcelona, Porta)

Impact level: short-term = demonstrated early (simulation, 33 neighbourhoods); medium and long term = expected.